

Vanshika Shah

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EDUCATION

Georgia Institute of Technology, Atlanta, GA, USA

Master of Science in Computational Science and Engineering (GPA: 4.0/4.0)

Aug 2024 - May 2026

(Key Focus: Machine Learning, Data Science, Generative AI, Data Analytics, Linear Algebra)

Vellore Institute of Technology, Chennai, TN, India

Bachelor of Technology in Electronics and Communication Engineering (GPA: 3.78/4.0 – Rank 10 out of 233)

Jul 2017 - Jun 2021

WORK EXPERIENCE

Graduate Teaching Assistant, Georgia Institute of Technology, Atlanta, GA, USA

Jan 2025 - May 2026

- **Machine Learning (Graduate level class):** Supported instruction on **supervised and unsupervised learning, model evaluation, optimization, and algorithmic implementations**, while reinforcing core ML concepts through assignments and discussion.
- **Computing for Data Analysis (Advanced Undergraduate Class):** Developed and demonstrated course materials and case studies on **statistical modeling, regression, and large language models**, integrating machine learning concepts to build a strong foundation.

Data Analytics Senior, Deloitte US-India, Hyderabad, TG, India

Aug 2021 - Jun 2024

- **Led a team of analysts** in large-scale financial data audits for **30+ US-based enterprises**, ensuring compliance and governance standards. Responsibilities included **training and supervising new analysts** as well.
- Evaluated alternative reporting and audit analytics approaches using **A/B testing methodologies**, improving process efficiency and insight generation.
- Built **predictive regression models** and implemented **anomaly detection algorithms** to identify inconsistencies in asset depreciation, reducing manual effort by 80%. Automated processing of trial balance data **at scale** using Python, SAS, and R, reducing execution time by 70%.
- **SQL-driven large scale data analysis** to support executive summaries and audit presentations. Also conducted **demographic and market trend analysis** across 20+ clients to support strategic policy decisions.
- Awarded **spot and applause awards** for technical expertise, leadership, and innovation in workflow automation.

Data Analytics Intern, Deloitte US-India, Hyderabad, TG, India

Dec 2020 - Feb 2021

- Used **SAS, SQL and Excel (VLOOKUPS, PivotTables)** to analyze, reconcile, and validate financial datasets (~300M lines), aligning with audit protocols. Additionally, used variance detection to flag high risk transactions. Used **Trifacta** to transform client data.

Research Intern, Georgia Institute of Technology, Atlanta, GA, USA (Supervisor: Dr. Omer Inan)

May 2019 - Jul 2019

- **Conducted biomechanical modeling research** utilizing **OpenSim** to simulate knee joint contact loads under varying gait conditions, aiding in osteoarthritis prevention research. Modelled physiological signal responses, **applying regression and optimization**.

ACADEMIC PROJECTS

Machine Learning-Based Gene Regulatory Network Reconstruction, Georgia Institute of Technology, GA, USA

Spring 2026

- Built and evaluated **Graph Neural Networks (GNNs)** and **tree-based ML models** for gene regulatory network reconstruction from **single-cell sequencing data**, leveraging **PyTorch Geometric, link prediction, and graph-based representation learning**.

Hybrid Predictive-Causal Framework for Mental Health Outcomes, Georgia Institute of Technology, GA, USA

Fall 2025

IEEE BHI 2025 Data Competition Winner

- Developed a **predictive causal framework** for mindfulness therapy outcomes using **Gaussian Copula augmentation** and a **shallow MLP**, achieving ~80% accuracy on heterogenous dataset. Also applied **SHAP** based interpretability with **propensity score causal inference** to identify subgroup specific treatment responses and optimal therapy dosage.

AI-Driven Clinical Trial Matching System, Georgia Institute of Technology, GA, USA

Spring 2025

- Built a scalable **GenAI retrieval pipeline** for clinical trial matching by integrating **FHIR/OMOP CDM, NLP models, LLMs, and RAG workflows, ChromaDB/vector search**, enabling semantic search, interactive Q&A, and personalized eligibility screening.

TECHNICAL SKILLS

- **Programming & ML:** Python (NumPy, Pandas, SciPy, Scikit-learn, Tensor Flow, PyTorch, Statsmodels, Keras), Machine Learning, Generative Models, Embedding Models, Similarity Metrics, Large Language Models, R, SAS, SQL, MATLAB, Trifacta. Feature Engineering
- **Data Visualization & Business Intelligence:** Tableau, Power BI, Matplotlib, ggplot, Seaborn, Plotly, MS Office, Excel, SQL querying
- **Math and Algorithmic foundations:** Bayesian Statistics, Probability, Linear Algebra, Causal Inferencing